Manzil JEE - Practice (2025)

Work, Energy and Power

Most Important JEE PYQs

Single Correct Type Questions

1. Identify the correct statements from the following:

A. Work done by a person in lifting a bucket out of a well by means of a rope tied to the bucket is negative.

B. Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.

C. Work done by friction on a body sliding down an inclined plane is positive.

D. Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero.

E. Work done by the air resistance on an oscillating pendulum is negative.

Choose the correct answer from the options given below:

[29 Jan, 2023 (Shift-II)]

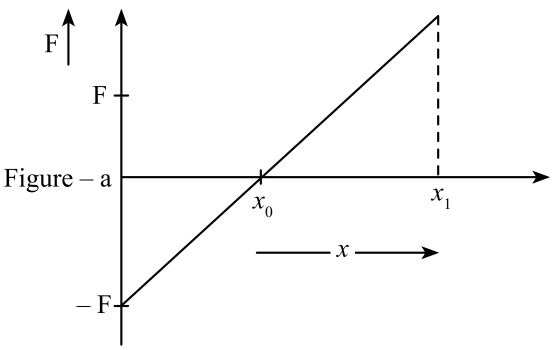
(*a*) B and E only (*b*) A and C only

(*c*) B, D and E only (*d*) B and D only

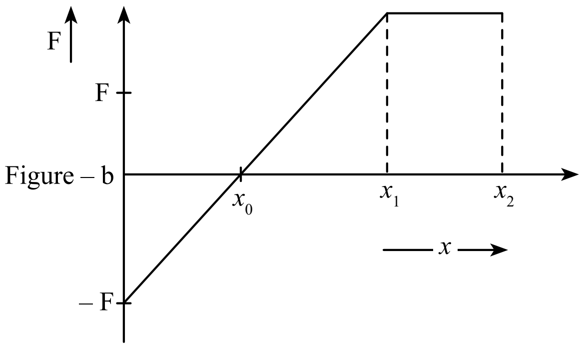
2. Arrange the four graphs in descending order of total work done; where W_1, W_2, W_3, and W_4, are the work done corresponding to figure a, b, c and d respectively.

[26 June, 2022 (Shift-II)]

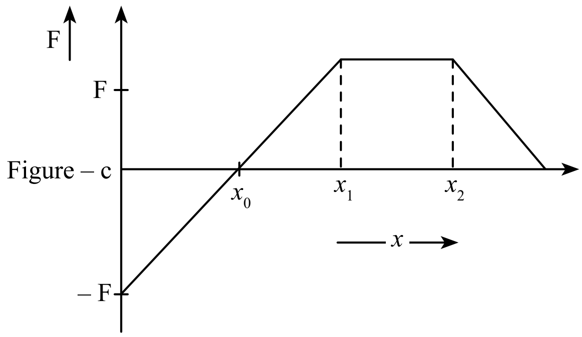
(A)



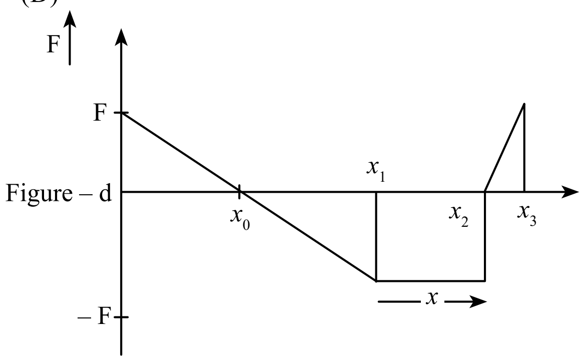
(B)



(C)



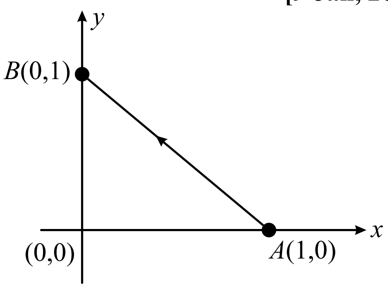
(D)



(*a*) $W_3 > W_2 > W_1 > W_4$ (*b*) $W_3 > W_2 > W_4 > W_1$

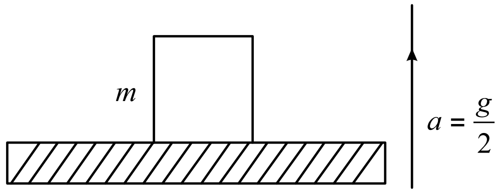
(*c*) $W_2 > W_3 > W_4 > W_1$ (*d*) $W_2 > W_3 > W_1 > W_4$

3. Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point $A(1,0)$ to $B(0,1)$ along the line segment is (All quantities are in SI units) **[9 Jan, 2020 (Shift-I)]**



(*a*) $\frac{1}{2}$ (*b*) $\frac{3}{2}$ (*c*) 2 (*d*) 1

4. A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in figure. Work done by normal reaction on block in time t is **[10 Jan, 2019 (Shift-I)]**



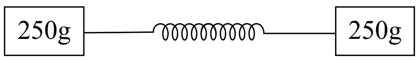
(*a*) $-\frac{mg^2t^2}{8}$ (*b*) $\frac{mg^2t^2}{8}$

(*c*) 0 (*d*) $\frac{3mg^2t^2}{8}$

5. A particle of mass 500 gm is moving in a straight line with velocity $v = bx^{5/2}$. The work done by the net force during its displacement from $x = 0$ to $x = 4$ m is (Take: $b = 0.25\ m^{-3/2}\ s^{-1}$) **[29 June, 2022 (Shift-I)]**

(*a*) 2 J (*b*) 4 J (*c*) 8 J (*d*) 16 J

6. As per the given figure, two blocks each of mass 250g are connected to a spring of spring constant $2Nm^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is: **[26 July, 2022 (Shift-I)]**



(*a*) $\frac{v}{2\sqrt{2}}$ (*b*) $\frac{v}{2}$

(*c*) $\frac{v}{4}$ (*d*) $\frac{v}{\sqrt{2}}$

7. A block of mass 2 kg moving on a horizontal surface with speed of 4 ms^{-1} enters a rough surface ranging from $x = 0.5$ m to $x = 1.5$ m. The retarding force in this range of rough surface is related to distance by $F = -kx$, where $k = 12\ Nm^{-1}$.The speed of the block as it just crosses the rough surface will be: **[28 June, 2022 (Shift-II)]**

(*a*) zero (*b*) 1.5 ms^{-1}

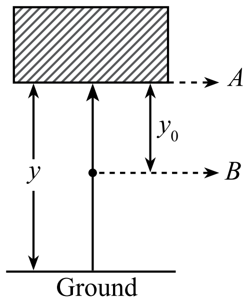
(*c*) 2.0 ms^{-1} (*d*) 2.5 ms^{-1}

8. A particle experiences a variable force $\vec{F} = (4x\hat{i} + 3y^2\hat{j})$ in a horizontal x–y plane. Assume distance in meters and force is Newton. If the particle moves from point (1, 2) to point (2, 3) in the x–y plane, then kinetic Energy changes by: **[24 June, 2022 (Shift-I)]**

(*a*) 50.0 J (*b*) 12.5 J

(*c*) 25.0 J (*d*) 0 J

9. In the given figure, the block of mass m is dropped from the point 'A'. The expression for kinetic energy of block when it reaches point 'B' is: **[29 June, 2022 (Shift-II)]**



(*a*) $\frac{1}{2}\text{mg}\,y_0^2$ (*b*) $\frac{1}{2}\text{mg}\,y^2$

(*c*) mg $(y - y_0)$ (*d*) mg y_0

10. A body of mass 'm' dropped from a height 'h' reaches the ground with a speed of $0.8\sqrt{gh}$. The value of work done by the air-friction is: **[1 Sep, 2021 (Shift-II)]**

(*a*) –0.68 mgh (*b*) 0.64 mgh

(*c*) mgh (*d*) 1.64 mgh

11. A particle which is experiencing a force, given by $\vec{F} = 3\hat{i} - 12\hat{j}$, undergoes a displacement of $\vec{d} = 4\hat{i}$. If the particle had a kinetic energy of 3 J at the beginning of the displacement, what is its kinetic energy at the end of the displacement? **[10 Jan, 2019 (Shift-II)]**

(*a*) 9 J (*b*) 12 J (*c*) 10 J (*d*) 15 J

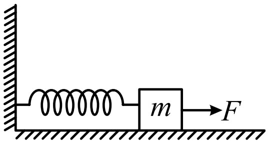
12. A wedge of mass M = 4m lies on a frictionless plane. A particle of mass m approaches the wedge with speed v. There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by:

129 April, 2019 (Shift-II)]

(*a*) $\frac{2v^2}{7g}$ (*b*) $\frac{v^2}{g}$ (*c*) $\frac{2v^2}{5g}$ (*d*) $\frac{v^2}{2g}$

13. A block of mass m, lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k. The other end of the spring is fixed, as shown in the figure. The block is initially at rest in an equilibrium position. If now the block is pulled with a constant force F, the maximum speed of the block is

[9 Jan, 2019 (Shift-I)]



(*a*) $\frac{2F}{\sqrt{mk}}$ (*b*) $\frac{F}{\pi\sqrt{mk}}$

(*c*) $\frac{\pi F}{\sqrt{mk}}$ (*d*) $\frac{F}{\sqrt{mk}}$

14. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $a = k^2rt^2$, where k is a constant.

The power delivered to the particle by the force acting on it is given as **[28 June, 2022 (Shift-I)]**

(*a*) Zero (*b*) $mk^2r^2t^2$

(*c*) mk^2r^2t (*d*) mk^2rt

15. A constant power delivering machine has towed a box, which was initially at rest, along a horizontal straight line. The distance moved by the box in time 't' is proportional to: **[18 March, 2021 (Shift-I)]**

(*a*) $t^{1/2}$ (*b*) t (*c*) $t^{3/2}$ (*d*) $t^{2/3}$

16. An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by :

[27 July, 2021 (Shift-II)]

(*a*) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

(*b*) $\left(\frac{9P}{8m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

(*c*) $\left(\frac{9m}{8P}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

(*d*) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}} t^{\frac{2}{3}}$

17. Potential energy as a function of r is given by, $U = \frac{A}{r^{10}} - \frac{B}{r^5}$, where r is the interatomic distance, A and B are positive constant. The equilibrium distance between the two atoms will be: **[24 June, 2022 (Shift-II)]**

(*a*) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$ (*b*) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$

(*c*) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$ (*d*) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$

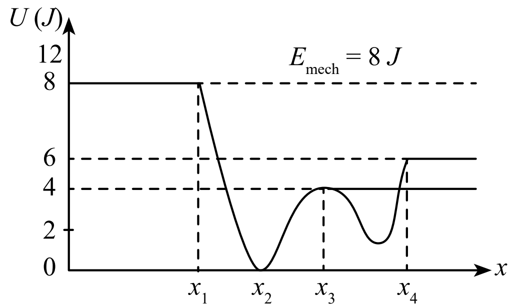
18. Given below is the plot of a potential energy function $U(x)$ for a system, in which a particle is in one dimensional motion, while a conservation force $F(x)$ acts on it. Suppose that $E_{mech} = 8$ J, the incorrect statement for this system is: **[27 July, 2021 (Shift-II)]**

(*a*) at $x = x_3$, *K.E.* = 4 J.

(*b*) at $x > x_4$, *K.E.* is constant throughout the region .

(*c*) at $x < x_1$, *K.E.* is smallest and the particle is moving at the slowest speed.

(*d*) at $x = x_2$, *K.E.* is greatest and the particle is moving at the fastest speed.



[where *K.E.* = kinetic energy]

Integer Types Questions

19 A force $F = (5 + 3y^2)$ acts on a particle in the y direction, where F is newton and y is in meter. The work done by the force during a displacement from y = $2m$ to $y = 5m$ is __________. **[1 Feb, 2023 (Shift-II)]**

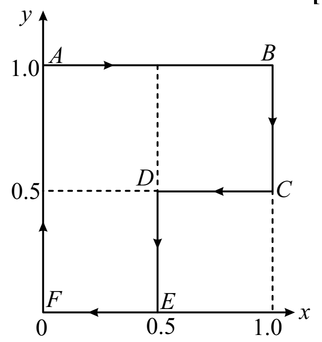
20. A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ m from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ m under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}$ N. The value of work done will be _____J.

[1 Feb, 2023 (Shift-I)]

21. A force $\vec{F} = (2 + 3x)\hat{i}$ acts on a particle in the x direction where F is in newton and x is in meter. The work done by this force during a displacement from $x = 0$ to $x = 4$ m, is _________ J. **[11 April, 2023 (Shift-I)]**

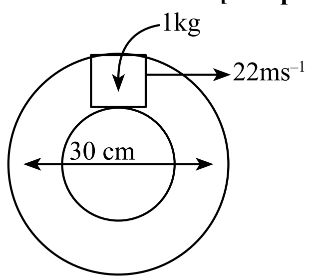
22. A particle is moved along a path AB-BC-CD-DE-EF-FA, as shown in figure, in presence of a force $\vec{F} = (\alpha y\hat{i} + 2\alpha x\hat{j})$ N where x and y are in meter and $a = -1$ N/m^{-1}. The work done on the particle by this force $\vec{F}$ will be ____ Joule

[JEE Adv, 2019]



23. A closed circular tube of average radius 15 cm, whose inner walls are rough, is kept in vertical plane. A block of mass 1 kg just fits inside the tube. The speed of block is 22 m/s, when it is introduced at the top of tube. After completing five oscillations, the block stops at the bottom region of tube. The work done by the tube on the block ________ is J. [Given g = 10 m/s^2]

[10 April, 2023 (Shift-I)]

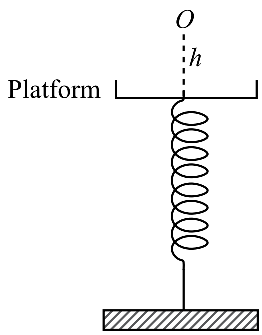


24. A body of mass 2 kg is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in 4s is $\frac{1}{3}\alpha^2\sqrt{P}$ m . The value of α will be **[30 Jan, 2023 (Shift-II)]**

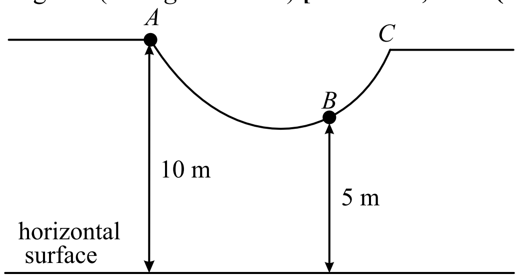
25. A body is dropped on ground from a height 'h_1' and after hitting the ground, it rebounds to a height 'h_2' If the ratio of velocities of the body just before and after hitting ground is 4, then percentage loss in kinetic energy of the body is $\frac{x}{4}$. The value of x is **[6 April, 2023 (Shift-II)]**

26. A ball of mass 100 g is dropped from a height h = 10 cm on a platform fixed at the top of a vertical spring (as shown in figure). The ball stays on the platform and the platform is depressed by a distance h/2. The spring constant is _______Nm^{-1}. (Use g = 10 ms^{-2})

[24 June, 2022 (Shift-I)]

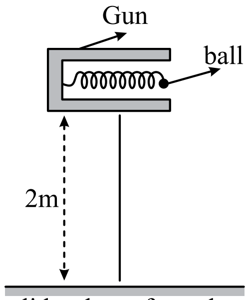


27. As shown in the figure, a particle of mass 10 kg is placed at a point A. When the particle is slightly displaced to its right, it starts moving and reaches the point B. The speed of the particle at B is x m/s. The value of 'x' to the nearest integer is (Take g = 10 m/s^2) **[18 March, 2021 (Shift-I)]**

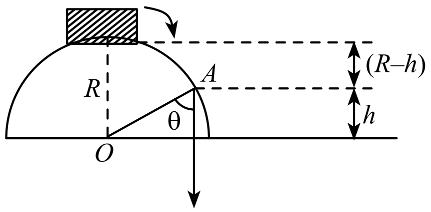


28. In a spring gun having spring constant 100N/m a small ball 'B' of mass 100 g is put in its barrel (as shown in figure) by compressing the spring through 0.05 m. There should be a box placed at a distance 'd' on the ground so that the ball falls in it. If the ball leaves the gun horizontally at a height of 2 m above the ground . The value of d is ___________m.

[20 July, 2021 (Shift-I)]



29. A small block slides down from the top of hemisphere of radius R = 3 m as shown in the figure. The height 'h' at which the block will lose contact with the surface of the sphere is _______ m. **[27 July, 2021 (Shift-II)]**
(Assume there is no friction between the block and the hemisphere)



30. An engine is attached to a wagon through a shock absorber of length 1.5 m. The system with a total mass of 40,000 kg is moving with a speed of 72 kmh^{-1} when the brakes are applied to bring it to rest. In the process of the system being brought to rest, the spring of the shock absorber gets compressed by 1.0 m. If 90% of energy of the wagon is lost due to friction, the spring constant is __________ $\times$ 10^5 N/m. **[1 Sep, 2021 (Shift-II)]**